Thai Travel Assistant

Capstone: Building LLM-Powered Applications (KMITL) · model `gemma-4-E4B-it` + `bge-m3` via the class proxy

1. Problem and users

Backpackers and students planning trips in Thailand ask concrete, checkable questions: what a ferry costs, how long a bus takes, how far an attraction is from town, and what a group will pay in total. General chat models answer these confidently from memory, with numbers that are often outdated or invented. Travel guides contain the answers but are long. For example, the Chiang Mai guide alone is about 157k characters.

This app answers such questions from **Wikivoyage guides for 15 destinations**, with a citation to the section each fact came from. It does arithmetic with a calculator rather than in the model's head, and it **says so when the guides don't cover a question** instead of guessing.

The corpus (15 guides, 709,647 characters, 2,499 chunks at the default size) has nothing in common with the course reference app's corpus, which documents the course's own software stack.

2. Design

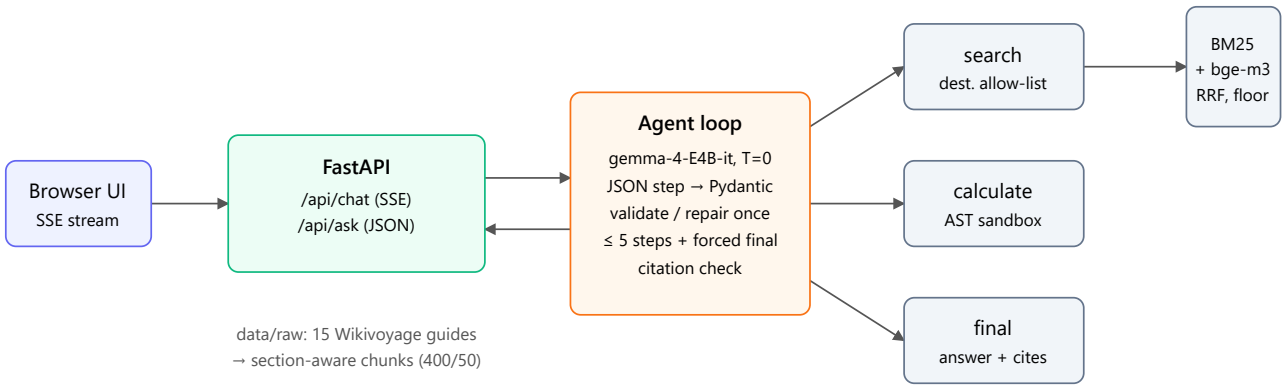


Figure 1: Architecture. Every model output is validated before anything runs.

Technique (course family)	How it is used	Why
Structured output + validation (1)	Each agent step is JSON parsed into a Pydantic discriminated union (search | calculate | final). An invalid step is sent back with the error once, then the agent gives up safely.	The proxy ignores `response_format`, and a 4B model often wraps JSON in prose. Nothing executes unless it validates.
RAG with citations (2)	Section-aware chunks with a "Destination — Section:" prefix, retrieved with hybrid BM25 + bge-m3 fused by RRF, and a cosine floor. Citations to chunks that were never retrieved are dropped.	Answers must be checkable. Hybrid retrieval catches both exact names and numbers (lexical) and paraphrases (dense).
Retrieval evaluation (3)	Hit@1, Recall@5 and MRR on 27 labelled questions, across 3 retrievers and 3 chunk sizes, plus a min_score sweep.	Every retrieval setting in the app is chosen from these numbers.
Agent loop with tools (4)	Tools: search (with an allow-listed destination filter) and calculate. The loop stops on `final` or after 5 steps, followed by one forced final turn. Repeated identical actions are refused.	Questions combining a fact with arithmetic need several steps. The destination filter is a form of query construction.
Sandboxed code execution (5)	The calculator parses expressions into an AST and evaluates only numeric literals and arithmetic operators. It	Small models make arithmetic mistakes, and running model-written code directly would be

Technique (course family)	How it is used	Why
	never calls `eval`, and exponents are capped.	unsafe.
Served interface (6)	FastAPI with `/api/chat` streaming each agent step over SSE, plus `/api/ask` returning validated JSON.	Users can watch the agent search and calculate as it works.

Deliberately left out: text-to-SQL and MCP. The data is prose, not tables, and adding them would have meant more moving parts without a question in the test set that needs them. All configuration (key, base URL, models, retrieval settings) comes from environment variables, and every evaluated call uses `temperature=0`.

3. Evaluation set and metrics

Test set: 33 hand-written questions (`eval/testset.jsonl`): 22 single-fact lookups, 5 lookup-plus-arithmetic questions, and 6 unanswerable questions about things absent from the guides (for example a "Sukhothai metro", Ko Lipe, or casinos). A script confirmed that every answerable question's **evidence phrase** appears verbatim in its gold guide.

Relevance rule: a chunk is relevant if it comes from the gold destination and contains the evidence phrase. Because the labels don't depend on chunk IDs, the same test set scores every chunk size.

Metrics:

- Retrieval: Hit@1, Recall@5 and MRR@10.
- Answers: keyword accuracy, meaning all gold keywords are present with commas normalised. For unanswerable questions, the model must abstain to count as correct.
- An LLM-as-judge score, as a second opinion.
- Cited support: the answer is correct *and* cites a chunk containing the evidence.

Baselines:

- For retrieval: BM25 only.
- For answers: *closed-book* (no retrieval) and *single-shot RAG* (one retrieval with the raw question, no tools).

4. Results

4.1 Retrieval: retriever × chunk size (27 answerable questions)

chunk_size	retriever	hit@1	recall@5	mrr@10	misses@10
400	bm25	0.56	0.85	0.67	q06 c05
400	dense	0.74	1.00	0.86	-
400	**hybrid**	**0.78**	**1.00**	**0.85**	**-**
400	hybrid+destination filter	0.78	1.00	0.85	-
800	bm25	0.63	0.96	0.76	-
800	dense	0.70	1.00	0.82	-
800	hybrid	0.70	1.00	0.83	-
1500	bm25	0.70	0.93	0.80	-
1500	dense	0.63	0.89	0.75	-
1500	hybrid	0.70	1.00	0.82	-

Baseline vs. chosen. At chunk size 400, the BM25 baseline gets Recall@5 = 0.85 and MRR = 0.67. The app's retriever (**hybrid**, highlighted) gets Recall@5 = 1.00 and MRR = 0.85. BM25 alone misses questions phrased differently from the guide. For example, "How much does the minivan from Chiang Mai to Pai cost?" against the guide's "Minivans (150 baht)" (q06, c05).

Chunk size (hybrid): 400: Hit@1 0.78, MRR 0.85; 800: Hit@1 0.70, MRR 0.83; 1500: Hit@1 0.70, MRR 0.82. Smaller chunks rank the exact passage higher, and at 1500 characters dense retrieval drops to Recall@5 = 0.89 because one vector has to cover several topics. The first version of this app used 800. This table is why it now uses `CHUNK_SIZE=400`. Dense and hybrid search are within one question of each other at 400 (MRR 0.858 vs 0.852). We kept hybrid because it has the best Hit@1, and BM25 helps with exact names and numbers.

Query construction. Filtering to the correct destination (the agent's `destination` argument) at size 400 adds nothing measurable here: every relevant chunk already ranks in the top 5. Its benefit is untested on this set, which has no questions where two guides compete.

4.2 Measured improvement: the min_score floor

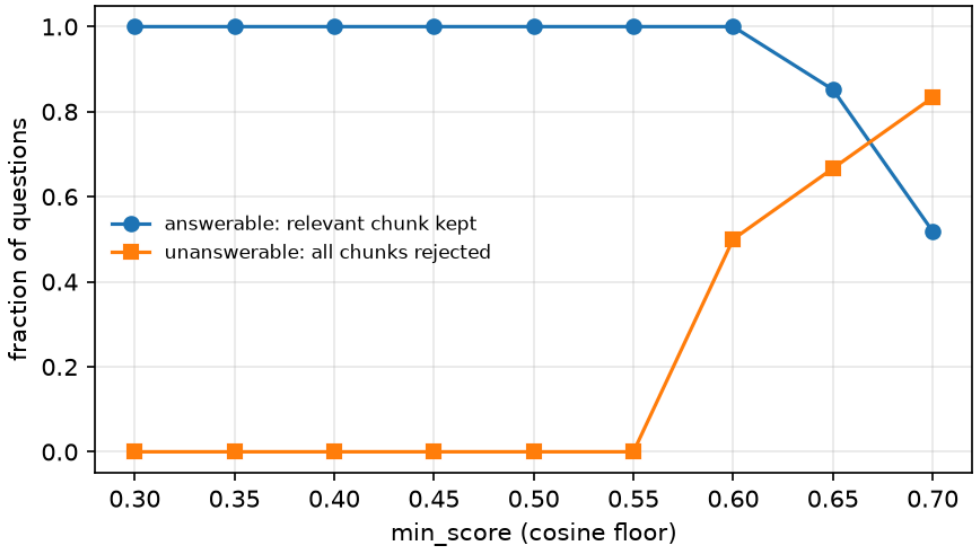


Figure 2: The min_score trade-off. A higher floor rejects more unanswerable questions but also drops real evidence.

min_score	answerable_kept	unanswerable_rejected	balanced
0.30	1.00	0.00	0.50
0.35	1.00	0.00	0.50
0.40	1.00	0.00	0.50
0.45	1.00	0.00	0.50
0.50	1.00	0.00	0.50
0.55	1.00	0.00	0.50
0.60	**1.00**	**0.50**	**0.75**
0.65	0.85	0.67	0.76
0.70	0.52	0.83	0.68

Naive default vs. chosen. With no floor (or any floor up to 0.55), retrieval always returns 5 chunks, so the model is handed unrelated text for **all 6** unanswerable questions and has to resist it on its own. Our rule was: *the highest floor that still keeps the evidence for every answerable question*. That gives **min_score = 0.60**, which rejects 0.50 of unanswerable questions while keeping 1.00 of answerable ones. 0.65 has a

marginally higher balanced score (0.76 vs 0.75), but it drops real evidence for 15% of answerable questions. For a travel app, a wrong "I don't know" is worse than letting the model see some junk it can still reject. With only 6 unanswerable questions, each one moves the rejection rate by 0.17, so this choice is coarse.

4.3 End-to-end answers (33 questions)

system	accuracy (all 33)	judge (all 33)	acc lookup	acc calc	acc unanswerable	cited_support (27)	false_abstain (27)	avg steps	parse_failures	avg sec
closed_book	0.27	0.42	0.18	0.20	0.67	0.00	0.22	1.00	0.00	0.80
single_shot	0.97	0.94	1.00	0.80	1.00	0.96	0.04	1.00	0.00	1.08
agent_v1	0.85	0.82	0.86	0.60	1.00	0.67	0.11	2.03	0.00	2.30
agent	**0.97**	**0.88**	**0.95**	**1.00**	**1.00**	**0.96**	**0.00**	**2.12**	**0.00**	**2.54**

Retrieval is what matters most. Closed-book gets 0.27: it invents plausible prices and distances, and it confidently answers 2 of the 6 unanswerable questions. Any grounded system scores above 0.85.

Agent vs. single-shot RAG. After the fixes in §5, the agent ties single-shot on overall accuracy (0.97 vs 0.97). It wins on arithmetic (1.00 vs 0.80), because single-shot has no calculator and sometimes refuses to multiply. It loses one lookup (q19, §5). It costs about 2× the latency (2.54 s vs 1.08 s) and about 2 model calls per question. On this test set, which is mostly single-hop questions, the agent's advantage is the calculator, not multi-step search.

agent_v1 → agent. The first agent scored only 0.85, with 0.11 false abstentions and 0.67 cited support. §5 explains the three fixes. **Caveat:** these fixes were found by reading failures on this same test set, so the agent's numbers are somewhat optimistic. A fresh held-out set is the main next step.

Metric reliability. The keyword metric and the LLM judge agree on 86% of 132 graded answers. Reading the disagreements, the judge (the same 4B model) mostly marks correct answers wrong when they include extra detail. Examples are q10, q21 and c03: the answers contain the right fact, and for c03 a correct calculation. We therefore treat keyword accuracy as the primary metric.

5. Failure analysis

5.1 What the first agent got wrong (agent_v1)

Each row is a question agent_v1 got wrong, or answered correctly without a valid citation:

id	question	gold	agent answer	searches	category
q02	How much does the river-crossing ferry cost in Ayutthaya?	5 baht	I couldn't find this in the travel guides.	river-crossing ferry cost	retrieval miss, then abstained
q05	Which famous Buddha image was once housed in a temple in Chiang Rai, and when?	The Emerald Buddha, in the 14th century	The Emerald Buddha, one of Thailand's most famous Buddha images, was once housed in Wat Phra Kaew in Chiang Rai during the 14th century. According to legend, the statue was found when lightning struck	famous Buddha image Chiang Rai temple	correct, but no valid citation
q12	How long is the van ride from Krabi Airport to Ko Lanta?	Roughly 2.5 hours	Scheduled van service from Krabi Airport to Ko Lanta takes roughly 2.5 hours, and a seat in a ten-person minivan from the airport to Ko Lanta costs about 300 baht [ko-lanta/get-in/13]. Another option	van ride Krabi Airport to Ko Lanta	correct, but no valid citation

id	question	gold	agent answer	searches	category
q15	At what time does the bat colony emerge near Khao Yai?	About 17:30	I couldn't find this in the travel guides.	bat colony emergence time	retrieval miss, then abstained
q18	How much does an SSI open water diving certification cost on Ko Tao?	Around 9,000 baht	The SSI open water certification on Ko Tao, which includes professional instruction, rental equipment, boat dives, and certification, is around 9,000 baht. Insurance and basic accommodation might also	SSI open water diving certification cost	correct, but no valid citation
q19	On which dates is Songkran (Thai new year) celebrated in Chiang Mai?	13-15 April	I couldn't find this in the travel guides.	Songkran dates	retrieval miss, then abstained
q20	How much does it cost to rent a motorbike per day on Ko Tao?	200-250 baht per day	Most bike shops on Ko Tao rent bikes for between 200 and 250 baht per day. It is advised to be very careful when renting from shops on the island. [ko-tao/stay-safe/6]	motorbike rental cost per day	correct, but no valid citation
q21	How many steps is the pilgrimage to the temple near Lopburi?	3,790 steps	There is a 3,790 step pilgrimage to the temple during the annual temple festival in Lopburi, during which the lights stay on 24/7. [lopburi/see/16, lopburi/do/1]	pilgrimage steps temple Lopburi	correct, but no valid citation
c01	Two people take the Lomprayah ferry to Ko Tao and back at the standard fare. How much do they pay in total?	750 x 2 people x 2 ways = 3,000 baht	The fare for a Lomprayah ferry to Ko Tao from Chumphon is 750 baht as of May 2022, and the fare from Maenam Pier is 700 baht (or 600 baht with a special promotion) as of May 2022. Assuming the standar	Lomprayah ferry fare Ko Tao	skipped the calculator
c04	How much do 3 people pay in total for a return longtail boat trip from Ao Nang to Railay?	(100 + 100) x 3 = 600 baht	The longtail boat charges 80 baht per person during the day to get to Rai Leh West and Ton Sai, and 100 baht per person after 18:00. For a return trip for 3 people during the day, the total cost would	longtail boat trip Ao Nang to Railay price	correct, but no valid citation
c05	What is the total minivan fare from Chiang Mai to Pai for a group of 4?	150 x 4 = 600 baht	As of September 2017, Prempracha offers minivans from Chiang Mai to Pai for 150 baht, which leave hourly from the Arcade terminal. The travel guides do not specify the fare for a group of 4.	minivan fare Chiang Mai to Pai	skipped the calculator

Reading these traces showed three distinct causes. Each fix is in the code and covered by a unit test:

1. **The floor emptied the agent's own searches** (q02, q15, q19). The floor was tuned on full questions, but the agent writes terse queries such as "Songkran dates". Chunks are embedded with a "Destination — Section:" prefix, so these queries scored around 0.52 cosine against a 0.60 floor. Every result was dropped, and the agent abstained. *Fix:* the search tool prefixes the destination onto the query, which raises the top cosine to about 0.65–0.70. The agent is also told to rephrase once before giving up.
2. **Citations in the wrong place** (q05, q18, q20, q21). The model wrote ids inline ("…250 baht [ko-tao/stay-safe/6]") or joined two ids in one string, leaving `citations` empty. *Fix:* ids are now also recovered from the answer text. Only ids that were actually retrieved in the run are accepted, so this cannot invent sources.
3. **Skipped calculator** (c01, c05). The model stated the unit price and stopped. *Fix:* a stricter prompt rule. A small model does not reliably follow "use the tool for arithmetic" unless the rule names the trigger (people, nights, days, return trips).

5.2 Remaining failures (current agent)

id	question	gold	agent answer	searches	category
q19	On which dates is Songkran (Thai new year) celebrated in Chiang Mai?	13-15 April	The temple of Wat Phra Singh in Chiang Mai is most attractive during Songkran, the Thai New Year, in mid-April [chiang-mai/see/41]. Additionally, the Songkran Festival (Thai Water Festival) takes plac	Songkran dates	answered without retrieved evidence

q19, a real retrieval miss. The chunk containing "13-15 Apr" is not in the top 5 for "Songkran dates". The agent found "mid-April" in a different chunk (about Wat Phra Singh), judged that good enough, and stopped searching. The answer is not wrong, but it is incomplete, and nothing in the loop checks that the answer is as specific as the question. A reranker, or a self-check step ("does the evidence answer the exact question?"), would target this.

An ambiguous gold label (q11 / c04). The Ao Nang guide gives the Railay longtail fare as both 100 baht (Understand) and 80 baht by day / 100 baht after 18:00 (Get in). Our label uses 100. Answers quoting 80 baht are arguably correct. We did not change the label after seeing results. It cost single-shot a judge point on q11, and it is why several answers hedge between the two fares. A labelling pass with two annotators would catch this kind of conflict.

6. Limitations and next steps

- **Tuned on the test set.** The chunk size, the floor and the agent fixes were all chosen by looking at these 33 questions. There was no separate dev/test split, so every number here is optimistic. Next step: write 30 fresh questions and run them once, untouched.
- **Small evaluation set.** With 33 questions (only 6 unanswerable), one question moves accuracy by about 3 points, so differences of one or two questions are within noise. Next step: grow the set to 100+ questions, including questions written by other students.
- **Stale facts.** Prices in the guides date from 2016–2025. The app reports what the guide says, not current prices. Next step: show each chunk's "updated" date next to its citation.
- **Single-hop bias.** Most questions need one passage. Questions comparing destinations, such as "cheaper to reach: Ko Tao or Ko Samet?", are untested.
- **Judge reliability.** The LLM judge is the same 4B model. A stronger judge, or a second human grader, would make that column more trustworthy.
- **English only.** The bge-m3 embeddings are multilingual, but Thai-language questions were not evaluated.

Reproduce: `python -m eval.eval_retrieval && python -m eval.eval_answers && python report/build_report.py`. Corpus text: Wikivoyage, CC BY-SA 4.0.